9.5.3 Storage class affects performance

The third factor that affects the performance of an EFS filesystem is the storage class used to persist data. By default, read latency to data persisted with standard storage class is as low as 600 microseconds. The latency for write requests is in the low single-digit milliseconds.

But if you choose the One Zone storage class to reduce costs, the read and write latency increases to double-digit milliseconds. Depending on your workload, this might have a significant effect on performance.

9.6 Backing up your data

AWS promises a durability of 99.999999999% (11 9s) over a given year for data stored on EFS. When using the Standard storage class, EFS replicates data among multiple data centers. Still, we highly recommend backing up the data stored on EFS, particularly so you can recover from a scenario where someone, maybe even you, accidentally deleted data from EFS.

Luckily, AWS provides a service to back up and restore data: AWS Backup. This service to centralize data protection supports EC2, EBS, S3, RDS, DynamoDB, EFS, and many more. The following three components are needed to create snapshots of an EFS filesystem with AWS Backup, as illustrated in figure 9.4:

- *Vault*—A container for grouping backups.
- *Plan*—Defines when and how to backup resources.